



Machine Learning for TSP Free-Acid Monitoring

Forecasting, virtual sensing, and an advisory OPC UA deployment
prototype

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Academic year: 2026–2027

June - September 2026

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List of Abbreviations

Abbreviation	Meaning
CSV	Comma-Separated Values
DCS	Distributed Control System
HMI	Human-Machine Interface
JITL	Just-In-Time Learning
KNN	K-Nearest Neighbours
MAE	Mean Absolute Error
ML	Machine Learning
OPC UA	Open Platform Communications Unified Architecture
OT	Operational Technology
PLC	Programmable Logic Controller
PLS	Partial Least Squares
RMSE	Root Mean Squared Error
TSP	Triple Super Phosphate
WAL	Write-Ahead Logging

1 Introduction

1.1 Context and motivation

Triple Super Phosphate (TSP) is a concentrated phosphate fertilizer obtained by reacting ground phosphate rock with phosphoric acid. In the TSP line studied during this internship, a slurry is formed, sprayed and dried, and its *free acid* content, recorded under the tag `SLURRY_FREE_ACID`, is the quantity the project tries to estimate and anticipate. The plant historian provides one-minute measurements of flows, temperatures, pressures and levels, but the free-acid value is a quality-type measurement that is not necessarily available as a continuous online signal.

Two operational questions follow. First, can the free acid be *estimated now* from process variables alone when no fresh measurement exists (virtual sensing)? Second, when a recent laboratory result *is* available, can its short-term movement be *anticipated* earlier than by waiting for the next result (forecasting)? These problems allow different inputs and must not be mixed: presenting a current estimate as an early warning, or a laboratory value as a live sensor, would overstate what any model can do.

1.2 Objectives

The general objective was a transparent, reproducible and strictly advisory prototype for free-acid monitoring. It was broken down into six specific objectives:

1. characterise the historian extract (cadence, gaps, extremes, variable semantics);
2. build a leakage-controlled, time-respecting feature contract (Chapter 4);
3. compare baselines and models under chronological validation (Chapters 5 and 6);
4. separate future forecasting from current-time virtual sensing;
5. assess robustness across chronological folds and select a model on stability, not only on mean error;
6. connect the saved artifacts to a read-only streaming architecture with a monitoring dashboard (Chapter 7).

1.3 Main contributions

The work produced (i) a documented feature contract of 178 engineered features per sample built from 21 sensors; (ii) a target-anchored delta forecast whose RMSE beats the persistence baseline by 19.8% to 34.9% on the held-out period (Table 6.1); (iii) a target-free virtual sensor with an empirical 90% interval and an out-of-envelope flag; (iv) a stability-oriented selection of a five-cluster Ridge model; and (v) a Docker/OPC UA replay prototype with data-quality shielding, SQLite audit storage and a native desktop dashboard.

1.4 Organisation of the report

Chapter 2 presents OCP Group, its activities, structure and professions. Chapters 3 to 7 describe the work carried out and its place in the plant's information flow: problem definition, data, methodology, results with interpretation, and the deployment prototype. Chapter 8 analyses limits and lessons, and Chapter 9 concludes and lists future work.

2 Company Presentation

2.1 OCP Group and its field of activity

OCP (Office Chérifien des Phosphates) was founded in 1920 and is a leading global producer of phosphate-based plant nutrition products, holding about 68% of the world's estimated phosphate reserves. Mining began at Khouribga in 1921 and extended to Youssoufia (1931) and Benguerir, while chemical production started at Safi (1965) and Jorf Lasfar (1984); a slurry pipeline linking Khouribga to Jorf Lasfar was inaugurated in 2014 [?]. For the first half of 2026 the group reported revenue of MAD 48.4 billion and an EBITDA margin of 28%, despite weaker fertilizer demand and a sharp rise in sulfur prices [?].

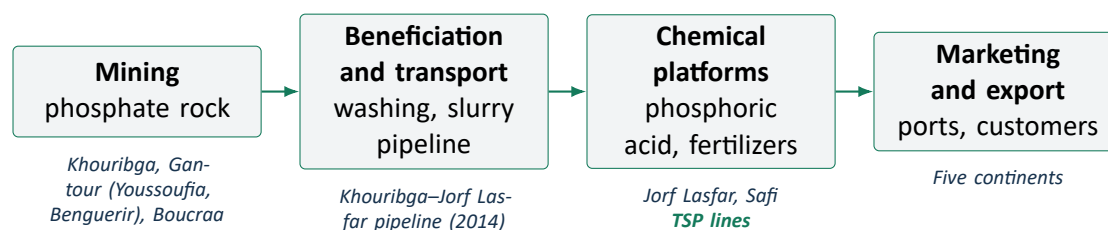


Figure 2.1. Simplified OCP value chain; the project belongs to the chemical stage.

Figure 2.1 places the internship in the chemical stage, where phosphoric acid and ground phosphate are converted into granular fertilizers. The data used by the project describe a TSP line: acid and phosphate feeds, reaction and passage tanks, slurry spraying, a dryer fed by fuel oil and hot air, and a recycle stream.

2.2 Organisational structure

OCP is organised along its value chain: mining sites, chemical platforms, and commercial and logistics activities, supported by transversal functions such as finance, human resources, maintenance and information systems. Figure 2.2 gives a simplified functional view of the units that interact with this project; it is a working representation built from the group's published value chain and the roles met during the internship, not the official organisation chart.

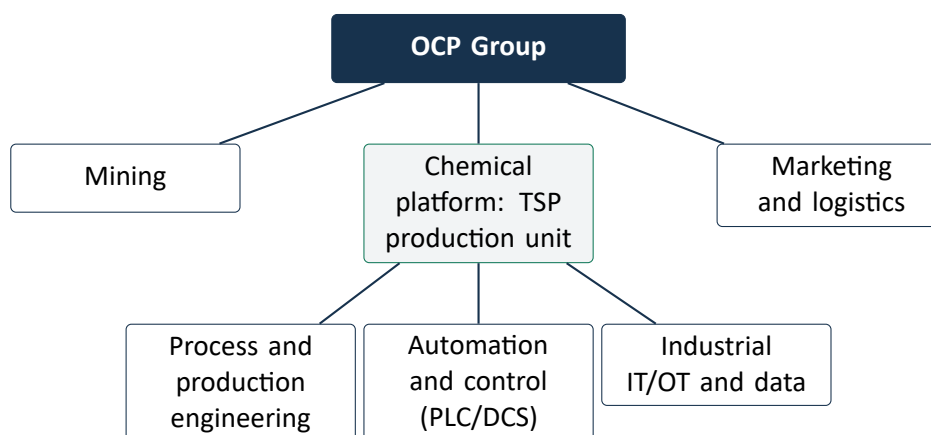


Figure 2.2. Simplified functional structure around the host production unit.

2.3 Professions involved in the project

The prototype sits at the intersection of several professions (Table 2.1), each of which is an interface of the design: process engineers define what the signals represent, automation engineers own the control layer that the prototype never touches, industrial IT/OT teams govern the network path, and data and software engineers maintain the models and the service.

Table 2.1. Professions met in the project and their link with the prototype.

Profession	Responsibility	Contribution expected by the project
Process and production	Operating conditions, product quality	Signal semantics, acceptable error, operating-event records
Automation and control	PLC/DCS logic, interlocks, tags	Tag list and Node IDs; read-only integration with no write path
Industrial IT and OT	Network zones, OPC UA endpoint, certificates	Gateway access, security policy, firewall route, host placement
Data and software engineering	Features, models, containers, storage, dashboard	Feature contract, inference service, tests, monitoring interface
Maintenance	Sensors and communications	Sensor health and communication status

3 Mission and Problem Definition

3.1 The TSP process and the target variable

TSP is produced by attacking ground phosphate rock with phosphoric acid; for fluorapatite the overall reaction is commonly written $\text{Ca}_5(\text{PO}_4)_3\text{F} + 7 \text{H}_3\text{PO}_4 + 5 \text{H}_2\text{O} \rightarrow 5 \text{Ca}(\text{H}_2\text{PO}_4)_2 \cdot \text{H}_2\text{O} + \text{HF}$ [?]. The free acid in the slurry is the acid that has not yet reacted, so it depends on the acid-to-rock ratio, temperatures and residence conditions. Figure 3.1 shows the simplified path reconstructed from the tag names of the data dictionary; the diagram is a simplified reading of the tag names and of the variable registry (Appendix A).

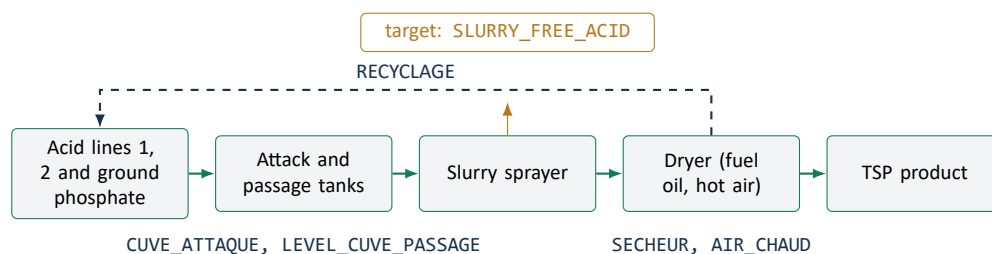


Figure 3.1. Simplified TSP line inferred from the data dictionary.

3.2 Three prediction contracts

The central methodological decision is to separate what each model is *allowed to know*. Let x_t be the process measurements at minute t , y_t the free acid, and $h \in \{1, 5, 10, 15\}$ the horizon in

minutes (Table 3.1). A fresh laboratory value y_t is a permissible input only for the target-anchored contract.

Table 3.1. Operational prediction contracts evaluated in the project.

Contract	Inputs	Output
Direct level	$x_{\leq t}$ only	$\hat{y}_{t+h} = f_h(x_{\leq t})$
Target anchored	$x_{\leq t}$ and a fresh y_t	$\hat{y}_{t+h} = y_t + \widehat{\Delta}y_{t,h}$
Target free	$x_{\leq t}$, no target history	$\hat{y}_t = g(x_{\leq t})$

3.3 Place of the work in the company

The prototype sits *beside* the control system, never inside it. It reads tags published through an OPC UA gateway, writes predictions to an audit database and shows them to engineers and operators on a dashboard. Table 3.2 lists what was delivered; the chain from historian data to advisory display is detailed in Chapter 7. Its design constraints are strict time ordering, sparse missing data, heterogeneous signal scales, a one-month history, and the requirement that a model or network failure must never affect plant control.

Table 3.2. Deliverables of the internship.

Deliverable	Content
Notebooks 01–08	Exploration, feature preparation, direct benchmark, delta and transition analysis, target-free virtual sensor, stability, regime-aware hybrid and JITL research
Saved artifacts	Feature registry, lag table, clipping bounds, model bundles (joblib) and result tables
Inference service	Python 3.11 service, OPC UA client, quality shield, rolling buffer, SQLite storage
Simulator	OPC UA server replaying the historian CSV (Stage 1)
Dashboard	PySide6 desktop application with trend, status cards and watchdog
Packaging and tests	Docker Compose, PyInstaller build, ZIP release, 19 unit tests

3.4 Evolution of the work

The work progressed from exploration to feature preparation, direct benchmarks, delta forecasting, a target-free virtual sensor, robustness research, and finally a replay-based integration prototype with a dashboard. The deployed target-free runtime keeps the five-cluster Ridge candidate for stability; the target-anchored Ridge delta model remains a separate configurable mode. Neither is a closed-loop controller: both produce advisory values only.

4 Data Understanding and Preparation

4.1 Data source and scope

The notebooks use a confidential historian export named `tsp_1min.csv`, absent from the repository. The generated audit artifacts record 44,640 one-minute rows from 1 to 31 January 2026, 21 numeric process measurements and the target, with 308 missing target values. Rows are sorted by the Date timestamp and every split preserves chronology (Table 4.1).

Table 4.1. Chronological split for the 1-minute horizon (rows and time span, UTC).

Split	Rows	Start	End
Train (70%)	31,011	2026-01-01 00:30	2026-01-22 17:17
Validation (15%)	6,645	2026-01-22 17:18	2026-01-27 08:58
Test (15%)	6,646	2026-01-27 08:59	2026-01-31 23:58

4.2 Variables and measurement semantics

The 21 inputs fall into flows (acid lines, ground phosphate, slurry, fuel oil, wash liquid, steam), thermal measurements (steam, attack tank, passage tank, sprayer, dryer outlet gas, hot air, brick), pressure and state signals (steam pressure, dryer depression, passage-tank level), and a recycle and production indicator. The variable registry (Appendix A) assigns each signal a role: *observed state*, *candidate actuator* (a flow, pressure or fuel setting that an operator may influence) or *uncertain role* (the recycle stream). The two slurry volumetric flows are expressed in m³/h. All other signals are modelled on a standardised scale, so that Ridge regression and tree ensembles are insensitive to the unit in which a tag is expressed; derived quantities are used as engineered indicators rather than as strict physical balances.

4.3 Exploratory data analysis

4.3.1 Structure and completeness

The extract contains 44,640 rows and 23 columns with no duplicated row or timestamp, which allows each row to be read as one distinct operating state. Missing values are rare (below 1% for every variable, Table 4.2), so rows were not dropped; instead missingness is encoded explicitly (Section 4.4).

Table 4.2. Variables with the most missing values (out of 44,640 rows).

Variable	Missing rows	Share (%)
DEPRESSURE_SECHEUR	444	0.995
FLOW_RATE_FIOUL	414	0.927
FLOW_RATE_VAPEUR	396	0.887
FLOW_RATE_PHOSPHORIC_ACID_2	393	0.880
SLURRY_FREE_ACID (target)	308	0.690

4.3.2 Distribution of the target

The target is tightly controlled around 11.3 with rare excursions (Table 4.3). The 99th percentile is 12.31, whereas the maximum reaches 52.67, giving a skewness of 12.1 and a kurtosis of 959. The Tukey fences $[Q_1 - 1.5 \text{ IQR}, Q_3 + 1.5 \text{ IQR}] = [10.05, 12.56]$ flag 352 rows, which occur both as isolated spikes (for instance 52.671 on 18 January, 25.964 on 19 January) and as continuous low-acid windows of 15 to 28 minutes (for example 28 January, 22:18–22:45, with values between 8.66 and 10.00). Because the distribution is heavy-tailed, the evaluation reports both MAE and RMSE and analyses extreme periods separately (Section 6.4); no target value is deleted automatically.

Table 4.3. Descriptive statistics of SLURRY_FREE_ACID (44,332 valid observations).

Statistic	Value	Statistic	Value
Mean	11.298	95 th percentile	12.012
Median	11.311	99 th percentile	12.308
Standard deviation	0.515	Minimum	3.392
1 st percentile	10.225	Maximum	52.671
25 th percentile	10.992	Skewness	12.08
75 th percentile	11.620	Kurtosis	959.0

4.3.3 Relationship with the process variables

Linear association with the target is moderate and uniformly positive (Table 4.4): no single signal exceeds a correlation of 0.37. Fuel-oil flow, steam flow, recycle, slurry flow and production load all rise with the free acid, which points to an interaction between feed intensity, heat load and internal circulation rather than to a single dominant driver. This observation motivated feature families that combine sources (ratios, gradients) instead of relying on raw values. Correlation is used for screening only.

Shifting a signal by a few minutes slightly increases its correlation (Table 4.4): the best lag is 1 to 5 minutes for most flows and 10 minutes for steam pressure and ground-phosphate flow. The gains are small, so no single dead time exists; the model therefore receives several lags and rolling statistics rather than one delayed copy (Section 4.5).

Table 4.4. Strongest Pearson correlations with the target and best lag.

Variable	r (lag 0)	Best lag (min)	r (best lag)
FLOW_RATE_FIOUL	0.364	1	0.356
FLOW_RATE_VAPEUR	0.359	3	0.361
RECYCLAGE	0.356	5	0.360
FLOW_RATE_SLURRY_T_PER_HR	0.346	1	0.353
PRODUCTION_TSP_BALANCE	0.339	2	0.340
PRESSURE_VAPEUR	0.323	10	0.327
FLOW_RATE_GROUND_PHOSPHATE	0.319	10	0.324

4.3.4 Physical plausibility

Several flow signals take small negative values (475 for steam flow, 451 for fuel oil, 420 for slurry volume, 354 for the production balance), consistent with noise around zero during low-flow or

stopped conditions, whereas 4,861 negative values of the dryer depression follow the draft sign convention. A single universal rule would therefore be wrong; the pipeline keeps the values, adds a missingness indicator and relies on percentile clipping of derived features.

4.4 Data quality and preprocessing

The findings of Section 4.3 translate into conservative rules. All fitted preprocessing is learned on the training period only. The first 30 rows are removed because the longest lookback is 30 minutes. Missing cells receive a missingness indicator and a training-median imputation. Derived ratios use the guarded form

$$r(a, b) = \frac{a}{b + \text{sgn}(b) \epsilon} \text{ if } |b| > \epsilon, \text{ otherwise missing, } \epsilon = 10^{-6}, \quad (4.1)$$

infinities are replaced by missing values, and process-derived features are clipped to the training 0.1th and 99.9th percentiles. Together these rules stop unstable ratios from dominating a model without fitting anything on validation or test data.

4.5 Feature engineering

For each source x and time t , the accepted temporal features are the current value, selected lags $x_{t-\ell}$, rolling means over $w \in \{10, 30\}$ minutes, a 10-minute rolling standard deviation, a five-minute difference $x_t - x_{t-5}$, and the least-squares slope over the last 10 samples,

$$b_t = \frac{\sum_{k=0}^9 (k - 4.5) (x_{t-9+k} - \bar{x}_t^{(10)})}{\sum_{k=0}^9 (k - 4.5)^2}, \quad (4.2)$$

which is set to missing if the window contains a missing value. Lags were chosen among $\{1, 5, 10, 15, 30, 60\}$ minutes from training-only correlation, mutual information, stability across chronological blocks and a limited process-response prior, keeping at most three lags per source; 20 of the 21 sources retained lags 1, 5 and 10.

Process-aware features add total acid flow, acid-flow imbalance $|F_{A1} - F_{A2}|$, the acid-to-phosphate ratio, total slurry volumetric flow, a slurry-density proxy $F_{\text{slurry}}^{t/h} / F_{\text{slurry}}^{m^3/h}$, temperature gradients (attack minus passage tank, hot air minus dryer outlet gas, steam minus sprayer slurry), a steam-energy proxy $P_{\text{steam}} T_{\text{steam}}$, a drying-intensity proxy, a recycle-to-production ratio and five 10-minute flow coefficients of variation $\sigma_{10} / (|\mu_{10}| + \epsilon)$. Only the two slurry volumetric flows are confirmed additive; every other relation is a proxy.

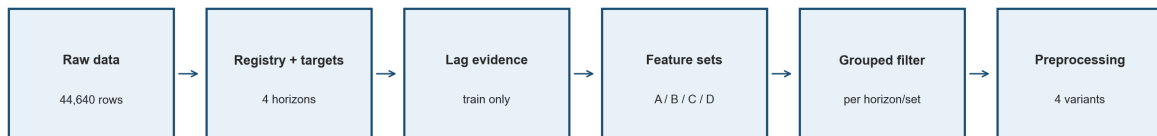


Figure 4.1. Leakage-controlled feature-preparation pipeline generated by Notebook 02.

Four hypotheses were tested (Table 4.5, Figure 4.1). Correlation filtering removed 48 temporal features whose training-only absolute correlation within a source family exceeded 0.98, while

protected process-aware features and missingness indicators were retained.

Table 4.5. Feature sets compared (counts are identical for all horizons).

Set	Hypothesis tested	Candidates	Retained	Missing flags
A raw	Current state alone is sufficient	21	21	0
B temporal	Lags and recent dynamics help	210	162	21
C process	Physical relations help	58	58	21
D full	Both are complementary	226	178	21

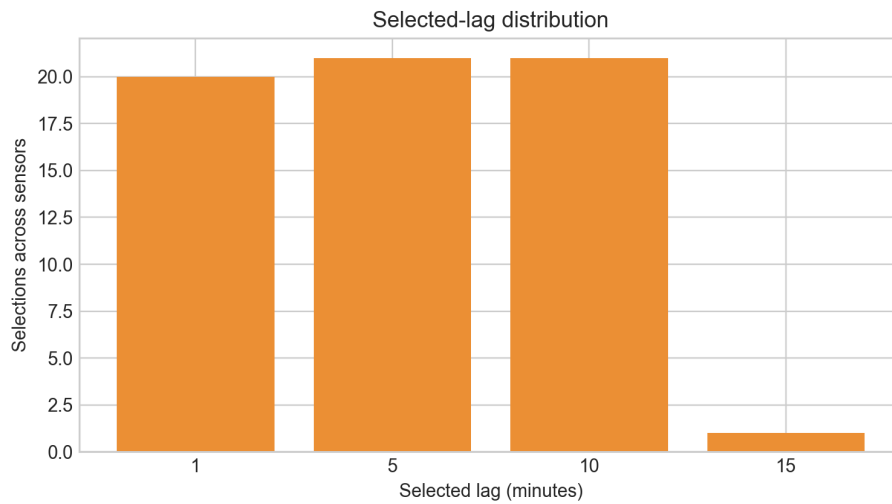


Figure 4.2. Distribution of the lags retained by the training-only selection procedure.

Figure 4.2 confirms that short lags (1, 5 and 10 minutes) dominate the selection, in agreement with the lag analysis of Table 4.4; the 15-minute lag appears for a single source (the attack-tank temperature) and the 30- and 60-minute lags are never retained.

5 Machine Learning Methodology

5.1 Problem formulation and metrics

Direct forecasting learns $f_h(x_{\leq t}) \approx y_{t+h}$. Because a measured level is very persistent over minutes, the project also evaluates delta forecasting,

$$\Delta y_{t,h} = y_{t+h} - y_t, \quad \hat{y}_{t+h} = y_t + \widehat{\Delta y_{t,h}}, \quad (5.1)$$

which concentrates the model on future *movement* and is valid only when the laboratory value is fresh. Target-free virtual sensing learns $g(x_{\leq t}) \approx y_t$ without any current, past or future target feature.

Errors are reported with MAE, RMSE and R^2 . Forecasts are compared with the operational persistence baseline $\hat{y}_{t+h} = y_t$ through the relative RMSE improvement

$$I_h = 100 \left(1 - \frac{\text{RMSE}_{\text{model}}}{\text{RMSE}_{\text{persistence}}} \right), \quad (5.2)$$

which is negative when the model is worse than persistence. RMSE weighs large errors more and highlights unstable periods; MAE is directly readable as an average deviation.

5.2 Chronological validation

The baseline split is 70%/15%/15% in time order (Table 4.1); shuffling is excluded because adjacent minutes are nearly identical states and would leak into the test set [?]. Lag selection, clipping bounds, correlation filtering, hyperparameters and blend weights were learned without test data. Stability research then used four expanding blocks of 1,657 validation rows each (Table 5.1); within each fold, imputation, scaling, clustering and local experts are fitted on the preceding rows only, giving a distribution of errors rather than a single number.

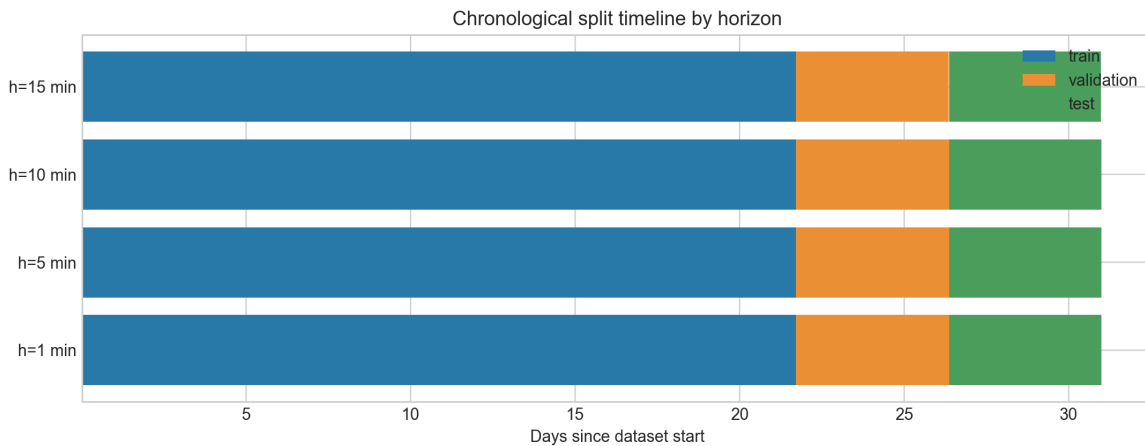


Figure 5.1. Chronological train/validation/test partition of the January series.

Figure 5.1 shows the partition; the validation and test periods always follow the training period in time.

Table 5.1. Expanding chronological folds used for stability analysis (UTC).

Fold	Train rows	Validation block
1	30,940	2026-01-22 17:18 to 2026-01-23 21:08
2	32,597	2026-01-23 21:09 to 2026-01-25 00:49
3	34,254	2026-01-25 00:50 to 2026-01-26 04:30
4	35,911	2026-01-26 04:31 to 2026-01-27 08:58

5.3 Models and tuning

Ridge regression, $\hat{\beta} = \arg \min_{\beta} \|y - X\beta\|_2^2 + \alpha \|\beta\|_2^2$ on standardised features, is the regularised linear reference for correlated engineered features [?]. Extra Trees [?] and histogram gradient boosting test nonlinear interactions; the target-free study adds PLS [?], KNN, XGBoost, LightGBM [?] and validation-weighted blends. Each family received a small, comparable search budget (two candidates per model, Table 5.2); the validation split selects the candidate.

The robustness notebooks compare global blends, KMeans-partitioned local Ridge experts (a cluster is assigned to each sample by nearest centroid in standardised feature space), a global/local hybrid and just-in-time learning (JITL) with PCA or PLS neighbourhoods [?]. KMeans clusters

Table 5.2. Search space of the target-free benchmark (Notebook 05).

Model	Candidate 1	Candidate 2
Ridge	$\alpha = 1$	$\alpha = 10$
PLS	5 components	10 components
KNN	$k = 25$, distance weights	$k = 75$, distance weights
Extra Trees	60 trees, leaf 2, features 0.8	60 trees, leaf 5, features 0.7
Hist. gradient boosting	80 iter., rate 0.05, 15 leaves	80 iter., rate 0.08, 31 leaves
XGBoost	150 trees, depth 4, rate 0.05	150 trees, depth 6, rate 0.05
LightGBM	150 trees, 15 leaves, rate 0.05	150 trees, 31 leaves, rate 0.05

are statistical groups of feature space, not confirmed operating regimes. The stability-first selection rule ranks candidates by mean RMSE, RMSE standard deviation across folds and worst-fold RMSE.

5.4 Uncertainty and drift flag

For the virtual sensor, an empirical 90% interval is $\hat{y}_t \pm q_{0.9}$, where $q_{0.9}$ is the 90th percentile of absolute validation residuals of the individual Ridge model (0.507). A drift flag marks samples lying outside the training envelope of the inputs. Both are advisory diagnostics, not guarantees.

6 Experimental Results and Model Selection

6.1 Forecasting: direct level versus target-anchored delta

Validation selected full-feature Ridge for the direct benchmark at every horizon, yet it did not beat persistence on the untouched test period: its RMSE was 23% to 68% worse (Table 6.1). This negative result was decisive, because it showed that reproducing the current level is easier than predicting its movement. Reformulating the task as the delta of Equation 5.1 with the hybrid feature set (188 features: 178 exogenous plus 10 target-history features) and $\alpha = 10$ reversed the outcome at all four horizons.

Table 6.1. Held-out test results (27–31 January): persistence, direct Ridge and target-anchored delta Ridge. I_h follows Equation 5.2.

h (min)	Persist. RMSE	Direct RMSE	Direct I_h (%)	Delta MAE	Delta RMSE	Delta R^2	Delta I_h (%)
1	0.197	0.332	−67.9	0.102	0.151	0.889	23.2
5	0.249	0.338	−35.1	0.113	0.169	0.862	32.3
10	0.280	0.344	−22.6	0.126	0.182	0.839	34.9
15	0.296	0.378	−27.5	0.176	0.237	0.726	19.8

The 10-minute horizon shows the largest relative gain and the 1-minute horizon the smallest absolute error; the operating horizon still depends on required reaction time and the cost of false or missed warnings. An ablation (Table 6.2) shows that exogenous process information

(178 features) and target history (10 features) are individually useful and complementary: their combination yields the best result at every horizon, while raw sensors alone barely improve on persistence.

Table 6.2. Ablation of Ridge delta models: RMSE improvement I_h (%) over persistence on the test period.

Feature set	Features	1 min	5 min	10 min	15 min
A raw sensors	21	0.5	2.7	6.2	1.7
D exogenous full	178	13.2	26.3	25.5	11.8
E target history	10	9.9	12.2	14.9	15.3
F hybrid (D + E)	188	23.2	32.3	34.9	19.8

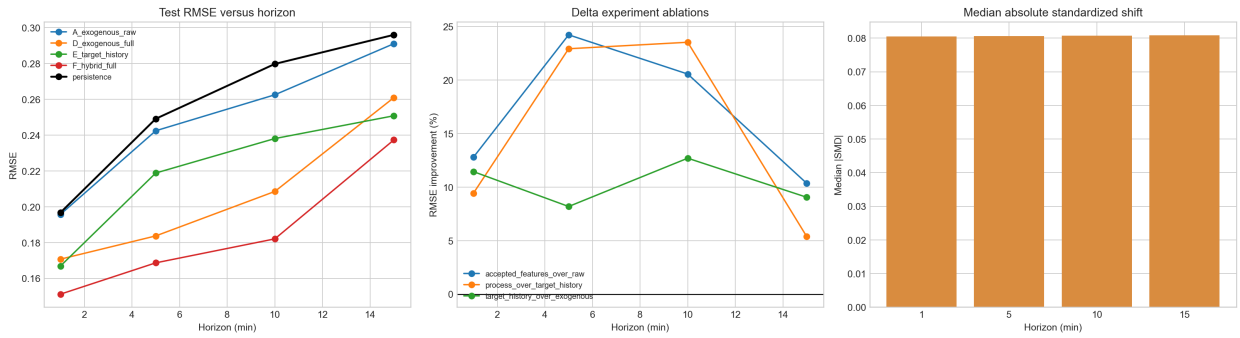


Figure 6.1. Delta forecasting performance, ablation and distribution-shift diagnostic.

Figure 6.1 combines performance, ablation and distribution shift. Transition weighting slightly reduced some local errors but worsened total RMSE and was dropped. A logistic transition classifier stayed diagnostic only: on the test period its precision at the 10-minute horizon was 0.161 for a recall of 0.512 and a false-warning rate of 0.267 (event prevalence 9.1%), too weak for an operational alarm.

6.2 Direct benchmark across feature sets

Before reformulating the task, the direct level benchmark compared four feature sets with three model families at each horizon. Table 6.3 gives the validation RMSE at the shortest and longest horizons. Engineered features help every family: moving from raw sensors to the full set reduces the Ridge RMSE from 0.375 to 0.335 at 1 minute, and the temporal set alone already captures most of this gain. Nonlinear models do not close the gap to the persistence baseline either, which indicates that the limitation lies in the formulation (predicting a level) and not in model capacity. Removing the 1% largest errors reduces the RMSE by 4 to 8% for these models, so extreme events inflate but do not explain the deficit.

6.3 Target-free virtual sensor

The target-free study excludes the current target, its history and future values. Table 6.4 shows that tree ensembles and boosting outperform Ridge on the test period, and that the validation-weighted 50/50 Ridge–LightGBM blend is best (MAE 0.249, RMSE 0.301, $R^2 = 0.558$) against RMSE 0.454 for the training-mean baseline. Its 90% interval covered 89.7% of test points with mean width 1.015; 91 of 6,632 test samples (1.4%) were flagged outside the training envelope, against none in validation. Because the interval is calibrated on individual-Ridge residuals while

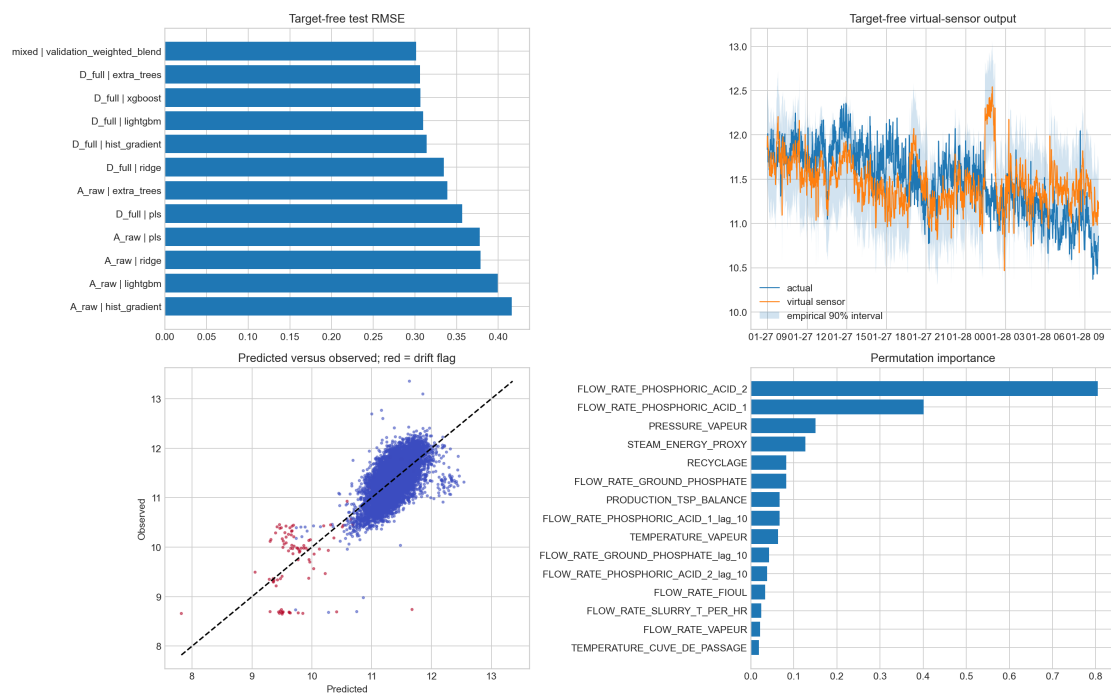
Table 6.3. Direct benchmark: validation RMSE by feature set and model.

Feature set	Model	1 min	15 min	Fit (s)	Infer. (s)
A raw	Ridge	0.375	0.395	0.03	0.004
A raw	Extra Trees	0.386	0.418	2.2	0.047
B temporal	Ridge	0.335	0.382	0.16	0.009
C process	Extra Trees	0.346	0.392	2.9	0.040
D full	Ridge	0.335	0.379	0.12	0.006
D full	Hist. gradient boosting	0.347	0.391	2.0	0.030

the point estimate is a blend, it remains an advisory diagnostic. Figure 6.2 shows the model ranking, the predicted series with its interval, predicted versus observed values with drift-flagged points, and the permutation importance discussed in Section 6.6.

Table 6.4. Target-free virtual sensor on the full feature set (178 features), test period.

Model	MAE	RMSE	R^2
Training mean (baseline)	0.355	0.454	0.000
Ridge ($\alpha = 10$)	0.273	0.335	0.455
Hist. gradient boosting	0.256	0.314	0.520
LightGBM	0.254	0.310	0.534
XGBoost	0.250	0.307	0.543
Extra Trees	0.246	0.306	0.544
Blend 50/50 Ridge–LightGBM	0.249	0.301	0.558

**Figure 6.2.** Target-free virtual-sensor comparison and diagnostics from Notebook 05.

6.4 Error analysis

6.4.1 Error distribution of the delta forecast

The delta forecast is accurate in the typical case and degrades gracefully in the tail (Table 6.5): the median absolute error is 0.08 at 1 minute and 0.14 at 15 minutes, the 95th percentile stays below 0.43, and the mean bias is within ± 0.03 , so the model neither systematically over- nor under-estimates.

Table 6.5. Test error distribution of the delta Ridge forecast (absolute errors).

h (min)	Median	P90	P95	Mean bias
1	0.080	0.202	0.252	−0.004
5	0.087	0.224	0.277	−0.005
10	0.099	0.250	0.304	−0.007
15	0.142	0.355	0.428	−0.023

6.4.2 Operating-regime analysis

To locate the errors, test samples were grouped by a multivariate normalised process-change score into stable, transition and extreme candidates (Table 6.6, 10-minute horizon). About 91% of samples are stable and are forecast with an RMSE of 0.147; the 306 extreme samples (4.6%) have an RMSE of 0.505, 3.4 times larger, and dominate the largest individual errors, which are cases where a sudden jump of 1.7 to 1.9 units occurs between consecutive minutes (for example on 31 January at 18:22). Such discontinuities are not anticipated by smoothly varying process signals and represent the main limit of a linear delta model. These groups are statistical categories of the sensor space, not labelled plant events.

Table 6.6. Test error by statistical operating regime, 10-minute delta forecast.

Regime	Samples	MAE	RMSE
Stable	6,027	0.114	0.147
Transition	297	0.155	0.190
Extreme	306	0.328	0.505

Figure 6.3 complements Table 6.6 with the behaviour of the transition classifier discussed above.

6.5 Stability research and model selection

Because January was reused during the experiment sequence, models were also compared across the four folds of Table 5.1 (Table 6.7). The 70/30 Ridge–LightGBM blend has the lowest mean RMSE (0.3191), but five-cluster Ridge is almost twice as steady (standard deviation 0.0396 against 0.0762) and has the best worst-fold RMSE of all candidates listed (0.3522 against 0.4236), so the runtime loads it for this stability-first reason. The 80% local and 20% global hybrid (0.3213, 0.0413, 0.3548) remained a research candidate, and JITL variants did not supersede the global blend.

Table 6.8 shows why the choice is not trivial: the blend is best in folds 2 and 3 but degrades sharply in fold 1 (RMSE 0.4236), whereas five-cluster Ridge never exceeds 0.3522. Fold 2 has a

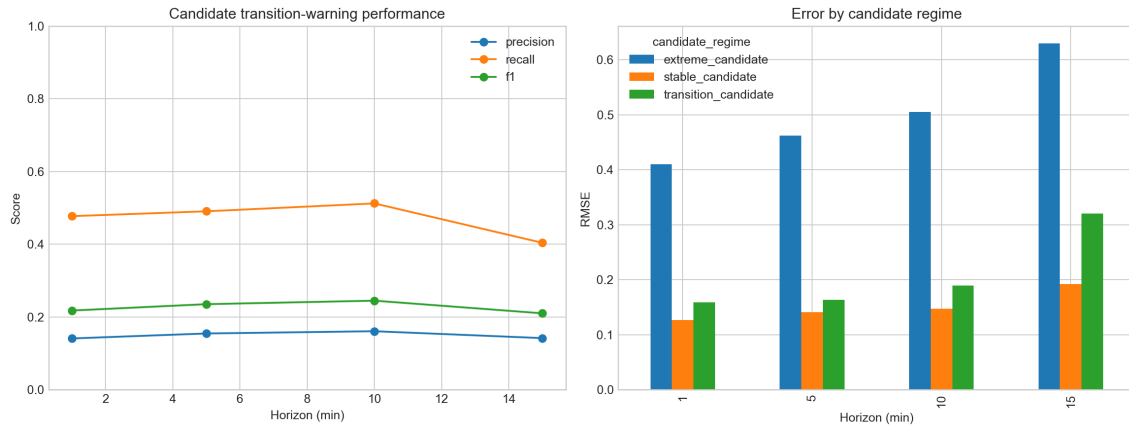


Figure 6.3. Warning-classifier behaviour and delta-forecast performance by operating regime.

Table 6.7. Four-fold stability comparison of target-free candidates (RMSE).

Candidate	Mean	Std. dev.	Worst fold
Blend 70/30 Ridge–LightGBM	0.3191	0.0762	0.4236
Blend 50/50 Ridge–LightGBM	0.3201	0.0686	0.4124
Global Ridge ($\alpha = 100$)	0.3274	0.0811	0.4353
Clustered Ridge, $k = 5$ (retained)	0.3280	0.0396	0.3522
Clustered Ridge, $k = 3$	0.3325	0.0494	0.3948
Hybrid 80% local / 20% global	0.3213	0.0413	0.3548

very low R^2 (0.018) for the clustered model, a reminder that within-block variance of the target is small and that RMSE and R^2 must be read together.

Table 6.8. Per-fold RMSE of the two leading target-free candidates.

Candidate	Fold 1	Fold 2	Fold 3	Fold 4
Blend 70/30 Ridge–LightGBM	0.4236	0.2846	0.2463	0.3217
Clustered Ridge, $k = 5$	0.3431	0.3522	0.2689	0.3477

Retrospective refits are reported as research outcomes and kept separate from the held-out results above.

6.5.1 Regime-aware hybrid and cost of the models

A later retrospective comparison blended the five-cluster local experts with the global model (80% local, 20% global). On the test period it reached an RMSE of 0.2903 against 0.2977 for the five-cluster Ridge (2.5% better) and 0.2985 for the global blend, with $R^2 = 0.590$ (Figure 6.4). Because January had already been consulted, this result is reported as a research outcome and is the first candidate for confirmation on new data. Table 6.9 adds the computational cost of the target-free models: Ridge trains in 0.11 s and predicts in 4 ms, boosting models train in 5 to 10 s, Extra Trees in 26 s, and the nearest-neighbour model is the slowest at inference (2.6 s for the test set) because it must search the training set for every prediction. All inference times are far below the one-minute cadence.

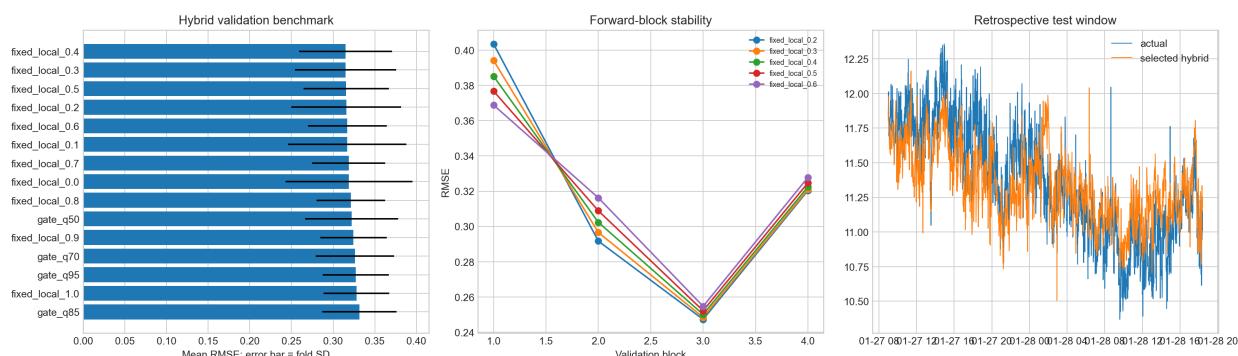


Figure 6.4. Regime-aware hybrid benchmark across local/global mixing weights and folds.

Table 6.9. Computational cost of target-free models on the full feature set.

Model	Train (s)	Test inference (s)	P95 abs. error
Ridge	0.11	0.004	0.601
PLS	0.92	0.016	0.629
KNN	0.06	2.576	0.792
LightGBM	5.45	0.053	0.559
XGBoost	9.53	0.059	0.556
Extra Trees	26.05	0.083	0.563

6.6 Model interpretability

No SHAP analysis was run; the notebooks use model-agnostic permutation importance (increase in validation mean squared error when a feature is shuffled), grouped importance by feature family, Ridge coefficients and importance-stability diagnostics, which is consistent with the known pitfalls of such methods [? ?]. Table 6.10 lists the most influential inputs of the target-free Ridge benchmark model.

Table 6.10. Largest permutation importances, target-free benchmark on the full feature set.

Rank	Feature	Family	MSE increase
1	FLOW_RATE_PHOSPHORIC_ACID_2	raw	0.805
2	FLOW_RATE_PHOSPHORIC_ACID_1	raw	0.401
3	PRESSURE_VAPEUR	raw	0.150
4	STEAM_ENERGY_PROXY	process-aware	0.127
5	RECYCLAGE	raw	0.082

By family, raw sensors account for most of the importance (1.77), followed by temporal features (0.28) and process-aware features (0.13); missingness indicators contribute nothing. The two acid flows dominate, which agrees with the chemistry of an acid attack on rock, but the result describes predictive association. It identifies which signals carry information about the free acid and is used to check that the model relies on plausible process drivers (acid feeds, steam and recycle), not to recommend changing a variable. The same caution applies to the multivariate normalised process-change score used in the notebooks and dashboard: it measures joint sensor movement, not target change, a confirmed startup or shutdown, or a model output.

7 Deployment and Implementation

7.1 Architecture and safety boundary

The implementation is a staged, advisory-only architecture (Figure 7.1). Sensors and laboratory results are handled by the PLC/DCS layer; Kepware is the intended industrial gateway, communicating southbound with approved plant protocols and exposing configured tags northbound through OPC UA [?]. A containerised Python service subscribes to these tags, builds features, predicts, and writes to a runtime volume holding the SQLite audit database and a health file. A native desktop dashboard on the host reads that volume read-only. The optional advisory OPC namespace is disabled by default (`advisory_write_enabled: false`).

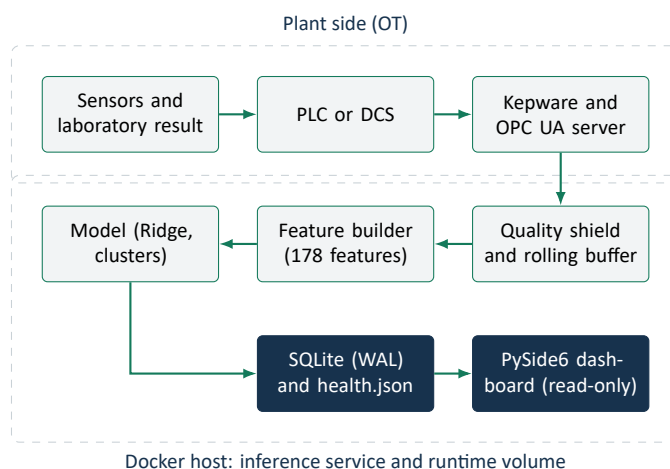


Figure 7.1. Advisory information flow implemented (Stage 1) and planned for plant integration.

The model has no path to PLC/DCS setpoints, controller modes, interlocks, permissives, valve or pump commands, or safety logic, and a service, network, database or dashboard failure leaves control unaffected. Plant integration follows the staged roadmap of Table 7.1. The laboratory configuration runs on an isolated Docker network with OPC security disabled (`security_mode: None`); the plant configuration uses the security profile of Table 7.2.

7.2 Three-stage deployment roadmap

Deployment is organised in three stages whose exit criteria are explicit (Table 7.1). Stage 1, the subject of this report, validates the software chain on replayed history; Stages 2 and 3 extend it to a laboratory gateway, then to read-only shadow operation and an advisory display. The OPC UA link itself is specified in Table 7.2: the inference service is the *client* and Kepware (or the simulator) the *server*, and the client subscribes to the sequence tag rather than polling every sensor.

7.3 Containers, configuration and storage

Docker Compose defines two services on an isolated bridge network: an OPC UA CSV-replay simulator and the inference service, both on Python 3.11 slim images. The inference image installs `libgomp1` for LightGBM, defines a health check and mounts configuration, model artifacts, schema and runtime output. Settings can be overridden through environment variables, and Node IDs live in `opc_tags.yaml`, not in code. Table 7.3 gives the main parameters.

Table 7.1. Staged deployment roadmap.

Stage	Environment	Scope and exit criterion
1	Docker, CSV replay through an OPC UA simulator	Tag mapping, subscriptions, snapshots, warm-up, quality policy, restart and reconnect, artifact loading, audit log (implemented)
2	Laboratory Kepware, then plant read-only shadow	Certificates, permissions, quality codes; predictions logged externally and compared with laboratory results; no effect on control
3	Advisory display (TSP.ML_Advisory namespace)	Predicted value, change, generation and target time, model status, input quality and model version, labelled <i>Advisory</i>

Table 7.2. OPC UA security profile for plant connection.

Requirement	Specification
Message security	SignAndEncrypt with trusted client and server certificates
Authentication	Anonymous access disabled; dedicated least-privilege service identity
Permissions	Read-only on plant input tags; writes only to a dedicated advisory namespace
Network	Firewall rule limited to the inference host and the approved OPC endpoint

Table 7.3. Main runtime parameters of the inference service (service.yaml).

Parameter	Value	Role
cadence_seconds (tolerance)	60 (5)	Expected source cadence and allowed jitter
maximum_buffer_rows	180	Rolling history kept in memory
heartbeat_interval_seconds	5	Health refresh, even without new data
subscription_period_ms	100	OPC UA subscription period
maximum_missing_required_share	0.25	Above this share, the snapshot is rejected
maximum_age_minutes (lab target)	120	Freshness limit for anchored mode

SQLite is the authoritative Stage 1 audit store, with JSONL and latest-JSON mirrors for inspection. It is opened in WAL mode with foreign keys and a 10-second busy timeout, so a read-only dashboard can query history while the service writes. Three tables (Table 7.4) make every output traceable to a run and a source timestamp; `UNIQUE(run_id, sequence)` prevents duplicate predictions. This suits a one-writer laboratory demonstration and is not a replacement for a plant historian, retention policy or backup design.

Table 7.4. SQLite data model.

Table	Main columns
service_runs	run_id, model version, mode, horizon, start/stop times, configuration snapshot (JSON)
predictions	run_id, sequence, source and prediction timestamps, predicted value, predicted change, input quality, imputed inputs, assigned statistical cluster
service_events	status transitions with time, sequence, message and details

7.4 Streaming pipeline and quality controls

Stage 1 replays the confidential CSV without plant access: the simulator publishes each row as OPC UA tags, sets the source timestamp and then writes a sequence number as a commit marker; it can refresh the simulated laboratory target every 60 rows. The service avoids mixed snapshots with a *sequence-read-sequence* protocol: it reads the sequence, then all input tags, then the sequence again, and accepts the row only if both sequence values agree.

Accepted snapshots pass the quality shield (Table 7.5). Isolated missing cells are tagged IMPUTED and handled by the saved training-fitted imputer and missingness indicators; nothing is ever fitted during live inference. The rolling buffer additionally refuses non-monotonic timestamps and clears its history when a source gap exceeds cadence plus tolerance (65 s), because rolling features must not silently bridge a gap.

Table 7.5. Data-quality rules applied to every snapshot.

Rule	Condition	Outcome
Missing inputs	$\leq 25\%$ of required tags missing	accepted, IMPUTED
Missing inputs	$> 25\%$ missing	DATA_ERROR
OPC quality	any required tag bad or uncertain	DATA_ERROR
Engineering limits	value outside configured minimum or maximum	DATA_ERROR
Chronology	non-monotonic timestamp or gap > 65 s	rejected or buffer reset
Laboratory target	missing, future-dated or older than 120 min (anchored mode)	DATA_ERROR

7.5 Warm-up and operating modes

The maximum feature lookback is 30 minutes, so a cold start needs the current row plus 30 previous rows: 31 accepted rows before the first valid prediction. Until then the service reports

WARMING_UP and does not fabricate history; historian backfill would be the production improvement. In `target_free` mode the service estimates the current value from process signals. In `target_anchored` mode it requires the current laboratory value with its timestamp and produces 1-, 5-, 10- or 15-minute delta forecasts, which protects it from treating an outdated laboratory result as a live analyser.

7.5.1 Processing sequence

Each minute the service executes the following steps; the complete flow is summarised in Figure 7.2.

1. Receive the sequence notification and read sequence, mapped inputs and sequence again.
2. Apply the quality shield (Table 7.5); isolated gaps become IMPUTED.
3. Append the accepted row to the chronological buffer; publish WARMING_UP until 31 rows exist.
4. Build the feature columns expected by the saved artifact and apply the training-fitted preprocessing, never refitting live.
5. Run `target_free` estimation or the configured `target_anchored` forecast.
6. Commit the record to SQLite in one transaction, then mirror it to JSONL and latest-JSON files.

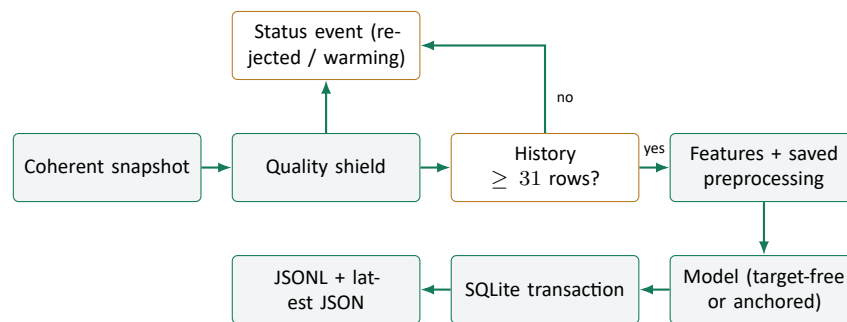


Figure 7.2. One-minute processing flow of the inference service.

7.6 Resilience, monitoring dashboard and delivery

The service refreshes its health file every five seconds, including when no new row arrives. An OPC sequence watchdog polls the sequence node during quiet periods to reveal a broken subscription or a gateway reset; connection closure or a sequence reset clears incompatible history, sets DISCONNECTED and starts bounded reconnection attempts. After recovery a new `run_id` is created, the buffer warms up again, and old-run predictions are never shown as current. The simulator waits for an acknowledgement after each sequence so that fast replay cannot skip rows; this flow control is laboratory-only and must be disabled with a real plant source. Table 7.6 lists the exposed states.

The monitoring surface is a native desktop application (PySide6 with a PyQtGraph timeline). It never blocks on I/O: a one-shot reader task runs on the Qt thread pool every two seconds, at most one at a time, and re-reads the trend history every fifth poll (about ten seconds). The SQLite file is opened with a `mode=ro` URI, `PRAGMA query_only` and a 500 ms busy timeout, and connections are closed deterministically to avoid Windows file-locking problems. The dashboard offers status and quality cards, a prediction timeline with selectable range, imputed-input markers, the change since the previous prediction, and a heartbeat watchdog whose threshold (30, 90 or 180 s, or off) separates a quiet process from a stale service.

Table 7.6. Operational states exposed by the inference service.

State	Meaning
STARTING or CONNECTED	Service started or OPC session open, no model-ready history
WARMING_UP	Accepted history is accumulating toward the 31-row requirement
RUNNING	A recent accepted snapshot produced a prediction
DATA_ERROR	A material quality or chronology rule blocked inference
DISCONNECTED	OPC communication failed or reset; reconnection is attempted

The timeline is interactive: hovering finds the nearest sample by binary search on the ascending timestamp array and draws a crosshair and an inspector with the exact time, value and whether the inputs were measured or imputed, and a click pins that point. The dashboard reads runtime output only and has no path to the plant.

For delivery on a Windows workstation, the dashboard is frozen with PyInstaller (`--onefile --noconsole`), a launcher script starts Docker Desktop if needed and brings up the services without rebuilding (loading an offline image archive when present), and a packaging script produces a single ZIP release. Six unit-test modules cover the rolling buffer, configuration, quality shield, storage, service resilience and dashboard data access. These tests exercise software behaviour only; they do not validate the chemistry, sensor calibration, industrial cybersecurity or the business usefulness of a prediction.

8 Discussion, Limits and Lessons Learned

8.1 Analysis of the results

The strongest evidence is the target-anchored delta formulation: on the held-out period it improved on persistence at all four horizons, whereas direct level prediction did not. Its use is conditional on a recent, correctly timestamped laboratory value, and the ablation shows that this value alone (set E) is not enough, as the gain comes from combining it with process information. The target-free virtual sensor answers a different question and must not be described as a future warning. The five-cluster Ridge decision favours repeatability over the lowest mean error; this is reasonable for a prototype meant to avoid erratic behaviour, but it is a selection made on one month, and the statistical clusters are groups of the sensor space that a future comparison with operating records (startups, shutdowns, maintenance) can interpret. Table 8.1 summarises the evidence level of each result.

Table 8.1. Evidence level and status of the main results.

Result	Evidence	Status
Delta Ridge forecast	Held-out test (27–31 Jan), beats persistence at 4 horizons	Main forecasting result
Target-free blend	Held-out test and 4 folds; 90% interval coverage 89.7%	Advisory diagnostic
Five-cluster Ridge	4 expanding folds, lowest variability	Retained in runtime
Hybrid and JITL	Fold comparison and retrospective test	Research results
Deployment chain	CSV replay, 19 automated tests	Stage 1 prototype

8.2 Scope of the study and perspectives

The study covers one continuous month (January 2026, 44,640 minutes) of a single production line, which is sufficient to characterise short-term dynamics and to compare models under strict chronological validation. Within this scope, two aspects frame the interpretation of the results. First, the free acid is fed to the pipeline as a laboratory-sourced tag, so the forecasting contract relies on a freshness rule (120 minutes in the service) and the virtual sensor serves the periods without a fresh value. Second, the models are predictive: feature importance describes association with the free acid and is not an intervention recommendation.

Uncertainty and drift indicators accompany each target-free estimate: the empirical 90% interval achieved 89.7% coverage and 1.4% of test samples fell outside the training envelope. The deployment chain was exercised end to end in a replay environment with 19 automated tests. The perspectives that follow naturally are

- (i) extending the evaluation to additional months with rolling-origin validation, using the frozen candidates of this report;
- (ii) confirming the regime-aware hybrid, which improved the RMSE by 2.5% retrospectively;
- (iii) relating the statistical clusters to the plant's operating records;
- (iv) running the service in read-only shadow mode against a secured gateway, as planned in Stage 2.

The architecture already contains the mechanisms this requires: the quality shield, the chronological buffer, warm-up, status events, heartbeat, reconnection and read-only storage.

8.3 Lessons learned

On the data-science side, the project showed why time ordering, a persistence baseline and leakage control are essential for one-minute process data, and that a change target can be more appropriate than an absolute level when the current value is highly persistent. On the software side, notebook logic became saved artifacts, a feature builder reproducing the accepted contract, configuration-driven OPC mapping, tests, containers and a read-only dashboard; industrial machine learning depends on traceability of feature meaning, inputs, health state and output history. The most important practical lesson is scope discipline: a simulator confirms integration behaviour, predictive importance describes association, and a prediction informs rather than commands the operator.

9 Conclusion and Future Work

This internship investigated free-acid monitoring for a TSP process from one month of one-minute data. The work moved from exploration and leakage-controlled feature engineering to benchmarks, target-anchored delta forecasting, target-free virtual sensing, fold-based stability research, and an advisory Docker/OPC UA prototype with a native dashboard.

The main forecasting result is a hybrid-feature Ridge delta model that beat persistence at 1, 5, 10 and 15 minutes, with RMSE improvements between 19.8% and 34.9% on the held-out period. A separate target-free Ridge–LightGBM virtual sensor outperformed the constant baseline (RMSE 0.301 against 0.454), and five-cluster Ridge was retained by the runtime for its stability across folds. These results answer different questions and have different evidence levels, so they are not combined (Table 9.1).

Future work follows the roadmap of Chapter 7. On the modelling side, it consists of evaluating the frozen candidates on further months with rolling-origin validation, confirming the regime-aware hybrid, calibrating the uncertainty on independent residuals and relating the statistical clusters to operating records. On the deployment side, the next milestones are a secured Kepware connection, historian backfill, a retention and backup policy and a longer read-only shadow operation with operator feedback (Stage 2), followed by the advisory display (Stage 3).

Table 9.1. Key results of the internship at a glance.

Component	Main result	Reference
Data and features	44,640 rows, 21 sensors, 178 leakage-controlled features	Chapter 4
Direct level forecast	23–68% worse than persistence	Table 6.1
Target-anchored delta Ridge	RMSE 0.151–0.237, i.e. 19.8–34.9% better than persistence	Table 6.1
Target-free virtual sensor	RMSE 0.301, $R^2 = 0.558$, 90% interval coverage 89.7%	Table 6.4
Stability-oriented model	Five-cluster Ridge: fold std 0.0396, worst fold 0.3522	Table 6.7
Regime-aware hybrid	RMSE 0.2903, 2.5% better retrospectively	Section 6.4
Deployment	Docker/OPC UA chain, SQLite audit, desktop dashboard, 19 tests	Chapter 7

Appendix A Evidence and Deployment Scope

A.1 Sources of evidence

The report is based on notebooks 01 to 08 of the repository, their generated CSV artifacts, the existing technical reports and the deployment source code. The artifacts remain the detailed audit record for all rows, folds, coefficients, predictions and configuration values summarised here.

A.2 Variable registry

Table A.1 lists the 22 registered variables. The two slurry volumetric flows are expressed in m³/h and all other signals are standardised before modelling; the role column follows the registry classification.

Table A.1. Variable registry (target in the last row).

Variable	Category	Role
FLOW_RATE_PHOSPHORIC_ACID_1, _2	flow	candidate actuator
FLOW_RATE_GROUND_PHOSPHATE	flow	candidate actuator
FLOW_RATE_FIOUL, _LIQUIDE_LAVAGE, _VAPEUR	flow	candidate actuator
PRESSURE_VAPEUR	pressure	candidate actuator
RECYCLAGE	production	uncertain role
PRODUCTION_TSP_BALANCE	production	observed state
FLOW_RATE_SLURRY_T_PER_HR	flow	observed state
FLOW_RATE_SLURRY_M3_HR, _M3_HR_2	flow	observed state (m ³ /h)
TEMPERATURE_VAPEUR, _CUVE_ATTAQUE, _CUVE_DE_PASSAGE	thermal	observed state
TEMPERATURE_SLURRY_PULVERISATEUR, _AIR_CHAUD	thermal	observed state
TEMPERATURE_GAZ_SORTIE_SECHEUR, _BRIQUE	thermal	observed state
DEPRESSURE_SECHEUR	pressure	observed state
LEVEL_CUVE_PASSAGE	level	observed state
SLURRY_FREE_ACID	quality	target, never an input

A.3 Software verification and reproducibility

The repository provides a pinned requirements.txt, two Docker images (simulator and inference service) with an explicit import lightgbm smoke test at image build, a launcher script, a PyInstaller build of the dashboard and a ZIP packaging script. Six unit-test modules (buffer, configuration, quality shield, storage, service resilience, dashboard data access) cover 19 test functions. All saved artifacts used at runtime (feature registry, lag table, clipping bounds, model bundles) are versioned with the code, and every prediction records the model version and run identifier in the audit database.

Machine Learning for TSP Free-Acid Monitoring

Final-year engineering internship report

Abstract. This report documents a machine-learning project for estimating and forecasting SLURRY_FREE_ACID from one-minute TSP process data. It presents leakage-controlled feature engineering, target-anchored delta forecasting, target-free virtual sensing, stability-oriented model selection, and an advisory-only Docker and OPC UA replay prototype. The study covers one month of one-minute data and a laboratory replay environment; its natural continuation is multi-month validation and read-only shadow operation on the plant network.

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